

1. Right triangle PYZ has vertices at $P(9,7)$, $Y(3,-1)$, $Z(9,-1)$. Right triangle RST has vertices at $R(6,-2)$, $S(1,4)$, $T(1,-2)$.

- a. Determine if $\triangle PYZ$ and $\triangle RST$ are congruent, similar, or neither.

$$PY = 10$$

$$YZ = 6$$

$$PZ = 8$$

$$RS = \sqrt{61}$$

$$ST = 6$$

$$TR = 5$$

neither

- b. What definition, theorem, or property justifies your conclusions?

Answers may vary. One possible answer: Since none of the sides in $\triangle RST$ are congruent to PY nor PZ , the shapes are not congruent based on definition of congruence. Since the pairs of corresponding sides are not proportional, the triangles are not similar.

- c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Using RST as the preimage, reflect $\triangle RST$ on $\overline{ST}$, then translate 8 to the right and 1 up, $(x,y) \rightarrow (x+7, y+1)$ to place T onto P . Mapping shows $\triangle RST$ and $\triangle PYZ$ are neither congruent nor similar.

2. Determine if the given polygons are congruent, similar, or neither.

- a. Trapezoid $FDGK$: $F(-8,-2)$, $D(-4,-2)$, $G(-2,-8)$, $K(-10,-6)$.
Trapezoid $MPRT$: $M(8,2)$, $P(4,2)$, $R(2,8)$, $K(10,6)$

☒ Congruent

☐ Similar

☐ Neither

- b. Kite $ABCD$: $A(-1,1)$, $B(4,1)$, $C(5,4)$, $D(2,5)$.
Kite $KFRP$: $K(0,0)$, $F(-1,-4)$, $R(-3,-3)$, $P(-4,-1)$.

- ☐ Congruent
☐ Similar
☒ Neither

- c. Rectangle $JKMP$: $J(-4,4)$, $K(0,-2)$, $M(3,0)$, $P(-1,6)$.
Rectangle $BCDF$: $B(8,6)$, $C(0,8)$, $D(-1,4)$, $F(7,2)$

- ☐ Congruent
☒ Similar
☐ Neither

3. Hannah and Zuki are determining if parallelograms $NPSE$ and $FGHR$ are congruent, similar, or neither. Parallelogram has vertices at $N(-9,2)$, $P(-1,5)$, $S(1,9)$, and $E(-7,6)$. Parallelogram $F(-4.5,1)$, $G(-0.5,2.5)$, $H(0.5,4.5)$, and $R(-3.5,3)$. Their work is shown.

Hannah	Zuki
Parallelograms $NPSE$ and $FGHR$ are similar because $\frac{NE}{FR} = \frac{PS}{GH}$	Parallelograms $NPSE$ and $FGHR$ are not similar because $\frac{NP}{FG} = \frac{SE}{HR}$

- a. Who do you agree with?
Hannah
- b. Explain the mistake that was made and how to correct it.
Zuki stated they are not similar and they are.